Sai Manish Reddy M

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Education

GITAM University — Hyderabad, India
Bachelor of Technology (B.Tech) – Computer Science — *2024 – Expected April 2028*
CGPA: 9.67/10.0

Relevant Coursework: Data Structures, Design & Analysis of Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Probability & Statistics, Linear Algebra, Calculus & Differential Equations

University of Toronto — Toronto, Canada
Studies in Mathematical & Physical Sciences — *2022 – 2024*
Honors: Dean's List Scholar (2023)

Relevant Coursework: Calculus with Proofs, Linear Algebra, Classical Mechanics, Quantum Physics, Programming

Experience & Leadership

G-ELECTRA Tech Club – GITAM University — Hyderabad, India
Web Development Lead — *August 2025 – August 2026*

- Led the development of two websites for GELECTRA, supporting tech-fest promotion, event information, and participant engagement.
- Collaborated with a team to integrate interactive 3D features and deliver responsive, production-ready web experiences.
- Optimized rendering performance, UI responsiveness, and cross-device compatibility.

Open Source Contributions

Matplotlib – Merged Pull Request #32543 — November 2026

- Fixed a misleading stable-version warning on development documentation pages by implementing a Sphinx `html-page-context` callback to hide the banner on `/devel/` pages.
- Integrated the callback into Sphinx's event system and verified the documentation build using `make html-noplot`.

Projects

NexCord | *Python, LangGraph, MCP, PostgreSQL, pgvector* — *September 2026 – October 2026*

- Built an agentic incident commander for live events that detects scheduling and resource conflicts, assesses incidents, and proposes actionable response plans.
- Designed an observe–analyze–approve–execute–verify–recover workflow that required human approval before consequential actions and verified outcomes after execution.
- Integrated LangGraph agents with a self-hosted MCP server and PostgreSQL/pgvector for tool execution, persistent state, semantic retrieval, and an auditable action history.
- Implemented recovery safeguards for failed actions, including rollback-oriented workflows to improve execution reliability.

SIFSentinel | *Python, scikit-learn, PyTorch, FastAPI, PostgreSQL, pgvector* — *August 2026 – September 2026*

- Built an AI/NLP-powered safety intelligence platform to identify potential Serious Injury and Fatality (SIF) precursors in incident, near-miss, and unsafe-condition reports.
- Implemented SIF classification, precursor extraction, IOGP Life-Saving Rule mapping, and risk-based prioritization to surface reports requiring attention.
- Integrated sentence embeddings and PostgreSQL/pgvector for semantic retrieval of similar historical reports and built an interactive safety analytics dashboard.
- Containerized the Next.js frontend, FastAPI backend, PostgreSQL database, and asynchronous ML worker using Docker Compose.

ClassSense_AI | *Python, MediaPipe, TensorFlow, Flask, Firestore, Gemini* — *January 2026 – March 2026*

- Developed a real-time classroom attendance system using multi-face detection and deep-learning face embeddings to recognize registered students and automate attendance logging.
- Integrated MediaPipe, TensorFlow/Keras, and Google Cloud Firestore through the Firebase Admin SDK for face processing and cloud-backed attendance records.
- Built a Flask dashboard for student registration, confidence-based attendance marking, searchable records, and Gemini-powered natural-language attendance analytics.

Technical Skills

Languages & Web	Python, C++, JavaScript, SQL, HTML, CSS, Next.js, Three.js, Streamlit
AI / ML & NLP	PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, OpenCV, MediaPipe, NumPy, Pandas, Hugging Face, Sentence Embeddings, RAG, Computer Vision, NLP
Agentic AI & LLMs	LangGraph, MCP (FastMCP), Amazon Bedrock (Nova), Gemini API
Backend & Data	FastAPI, Flask, Django, PostgreSQL, pgvector, Firestore
MLOps & Cloud	Git, Docker, Docker Compose, Kubernetes, Linux, MLflow, Google Cloud Platform, AWS
Core CS	Data Structures & Algorithms, Object-Oriented Programming, Operating Systems Basics, DBMS

Awards & Hackathons

Smart India Hackathon – Internal Round Qualifier — September 2026

- Qualified through the institutional internal round among **154 teams** with **SIFSentinel**, developed for Oil India Limited's problem statement 26165.

EduHacks – First Overall Winner — December 2022 · MLH

- Built **Slidesmart**, an AI tool that converts YouTube videos and documents into concise presentation slides.

NewHacks – Sponsor Prize Winner, Synopsys Smart Everything — November 2022 · MLH · University of Toronto

- Built **tPay**, a facial-recognition prototype for student identity verification and payment authentication.

Harvest Hacks – 2nd Place Overall — November 2022 · MLH

- Built the Streamlit frontend and integrated a transfer-learned ConvNeXt detector trained on Google Cloud TPUs to classify plants and return weed bounding boxes.